

# Syllabus

## Spring Semester 2026

|              |                                   |             |        |                     |                 |        |                     |           |                      |
|--------------|-----------------------------------|-------------|--------|---------------------|-----------------|--------|---------------------|-----------|----------------------|
| Course Title | Fundamentals of Organic Chemistry | Course Code |        | CB4102              |                 | Credit | 3                   | Classroom | Hyeongseol-gwan 3303 |
|              |                                   | Type        |        | elective            |                 | Hours  | 3                   |           |                      |
| Teacher      | Name                              | Class No.   | Office | Office Hour*        | Contact         |        | E-mail              |           |                      |
|              | Yeosan Lee (이여산)                  | 2           | 2505   | Tuesday 1-4 periods | +82-51-606-2223 |        | yeosanksa@ksa.hs.kr |           |                      |

\* Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

## 1. Course Description

The Fundamentals of Organic Chemistry course will cover the properties and reactivities of organic and organometallic compounds containing various functional groups. The course will address the detailed characteristics and reactivities of functional groups such as alkenes, alkynes, halides, alcohols, ethers, carbonyl groups, amines, and others. The analysis of organic and organometallic molecules will be performed from the perspective of physical organic chemistry, with a focus on detailed mechanistic explanations of their chemical reactivities. Additionally, a team project involving computational chemical analysis will be conducted to deepen understanding of organic and organometallic compounds and their associated reactivities.

## 2. Core Target Competencies

| Subject Core Competencies                                                                                                                                                                                                                                                                                                                                                                                                             |                       |                                       |                                |                          |            |                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------|--------------------------------|--------------------------|------------|------------------------|
| Analytical and Creative Thinking                                                                                                                                                                                                                                                                                                                                                                                                      | Complex Communication | Integrity and Ethical Decision Making | Complex Problem-solving Skills | Critical Thinking Skills | Creativity | Ability to Collaborate |
| Analytical and Creative Thinking / Complex Communication / Leadership and Teamwork / Global Perspective / Adaptability, Progression, and Spirit of Adventure / Integrity and Ethical Decision Making / Habits and Mindset / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate / Judgement and Decision-making Skills / Service Orientation / Negotiation Skills / Cognitive Flexibility |                       |                                       |                                |                          |            |                        |

## 3. Textbooks and References

### 3.1. Textbook

(1) Fundamentals of Organic Chemistry, 7th Edition, by John McMurry (Cengage Learning)

### 3.2. References

(1) Organic Chemistry, 9th Edition - by John McMurry (Cengage Learning)

(2) Inorganic Chemistry, 5th Edition - by Gary L. Miessler, Paul J. Fischer, and Donald A. Tarr (Pearson)

(3) Advanced Organic Chemistry, Part A: Structure and Mechanisms, 5th Edition - by Francis A. Carey and Richard J. Sundberg (Springer)

## **4. Class Method**

### **4.1 Class Management**

- (1) The lectures will be delivered in class using the board and/or presentation materials. Students are required to take handwritten notes in every class to enhance their summarization, clarification, and graphical description skills. The students' in-class notes will be submitted and assessed based on their content.
- (2) A problem set will be given to the students after finishing each chapter in class. Students are required to solve and submit the given problems to enhance their complex problem-solving skills. The students' problem set answers will be submitted and assessed based on their content.
- (3) Students are required to participate in discussions regarding the given topics and questions to enhance their communication and critical thinking skills. The students' participation will be assessed based on their content and activity.
- (4) Assessment will be performed based on both examination and non-examination criteria. Two exams (mid-term and final) will be conducted as part of the examination evaluation. One team project involving computational analysis, assignments, attitude, in-class participation, and attendance will be evaluated as part of the non-examination criteria.
- (5) All of the students' evaluation materials (e.g., lecture notes, problem sets, project outcomes, etc.) must be submitted to the Google Classroom.
- (6) Any late submission after the due date or time will not be accepted, resulting in 0 point for the related criteria.

### **4.2 Academic Integrity Policy**

- (1) The diversity of the class members ought to be respected. Any type of discrimination regarding ethnicity, nationality, language, personal orientation, etc., will not be tolerated.
- (2) All communications will be conducted in English as part of addressing the language diversity of all members. Other languages are not allowed unless permission is granted by the instructor.
- (3) Any type of misconduct (e.g., cheating, copying, possession of electronic devices during exam, refusing to follow instructions, dishonorable actions, other unauthorized activities not related to the course, etc.) will automatically result in an FAIL grade for the course. Furthermore, the student will receive official disciplinary punishment according to the Korea Science Academy Regulations.
- (4) Plagiarism and copying are strictly prohibited. Citations and references must be provided on a sentence-by-sentence basis whenever a source is used. Any sequence of six or more consecutive words taken from another source will be regarded as plagiarism. Note that paraphrasing is required whenever sources are referred to. All visual and technical materials (e.g., figures, equations, graphs, tables, and code) must be properly cited, and any reuse must comply with applicable permissions and licensing. Students must comply with all other regulations regarding academic integrity.
- (5) Using AI tools (e.g., ChatGPT) for academic support is permitted; however, students must verify and study the original sources cited or referenced. In addition, students must cross-check information using Google and multiple scholarly references. In this course, AI-generated content is not accepted as a citable source or reference. Submitting AI-generated content without proper citation to original sources will be regarded as plagiarism and unauthorized copying.
- (6) The use of AI tools (e.g., ChatGPT) to write or generate the contents of reports is strictly prohibited. The school may use plagiarism-detection tools (e.g., CopyKiller, Turnitin, GPTKiller, etc.). If a report exceeds a specified threshold (e.g., 15% similarity or higher), it will be considered plagiarism/unauthorized copying and noncompliance with the instructions.

## 5. Evaluation

### 5.1 Evaluation Criteria

| Mid-term Exam | Final Exam | Computational Chemistry Project | Lecture Note and Problem Set Assignments | Attitude and In-class Participation | Attendance | Sum |
|---------------|------------|---------------------------------|------------------------------------------|-------------------------------------|------------|-----|
| 30            | 30         | 15                              | 10                                       | 10                                  | 5          | 100 |

### 5.2 Evaluation Method

#### 5.2.1 Mid-term and Final Exam (30 points each; total 60 points)

- (1) One mid-term and one final exam will be held after covering specific chapters. Understanding basic knowledge, creative thinking, cognitive flexibility, literal explanation, and graphical demonstration will be important competencies for the evaluation.
- (2) Each exam will be held for approximately 180 minutes. The exam duration may be adjusted based on the given circumstances.
- (3) Each student is allowed to prepare 3 pages of Memory Sheets for the exam. The blank Memory Sheets will be provided to the students at the start of the course and right after the mid-term exam. The prepared Memory Sheets may be possessed by the student during the exam only if (a) they are approved by the instructor on the exam date, (b) they include the student's handwritten Honor Code, and (c) they bear the signatures of both the student and the instructor.
- (4) Along with the other criteria, the exams will be administered in strict accordance with the Korea Science Academy Regulations and the [4.2 Academic Integrity Policy] section.
- (5) The make-up score for an exam missed due to an excused absence or a medical absence shall be calculated as follows: determine the student's percentile rank based on the total score from other exams in the same course that the student has taken, assign the score on the missed exam corresponding to the same percentile, and then multiply it by 100% (excused absence) or 80% (medical absence). Further details are available in the Korea Science Academy Regulations.

#### 5.2.2 Computational Chemistry Project (15 points)

- (1) Each student will perform a individual computational analysis of organic/organometallic chemical phenomena. The theoretical background and program utilization will be covered in class.
- (2) Each student is required to discuss the topic and detailed computational methodology with the instructor before the project begins.
- (3) The project results will be summarized and submitted as a PDF report. The raw computational data will be submitted as text files. Each student will present their project results in class.
- (4) The content of submitted PDF files must be clear and easy to read. Any vague content in terms of light and dark contrast, color, etc., will not be assessed, resulting in 0 point for the related criteria. The use of a scanner is recommended over the use of a cell phone camera if necessary.
- (5) The total points for the Computational Chemistry Team Project is 15. The detailed assessment criteria are:
  - (a) setting up a creative and complex topic (3 points)
  - (b) computational method and progress (3 points)
  - (c) result analysis and discussion (3 points)
  - (d) report content (3 points)
  - (e) presentation skills (3 points)
- (6) Any late submission after the due date or time will not be accepted, resulting in 0 point for the related criteria.

### 5.2.3 Lecture Note and Problem Set Assignments (10 points)

- (1) Students are required to take handwritten notes in every class. Each set of notes must be submitted as a single PDF file. Taking handwriting notes on tablets (e.g., iPad, Galaxy Tab, etc.) is allowed. However, lecture notes created using word processing software (e.g., Microsoft Word, Pages, Hangeul, etc.) are not permitted.
- (2) Students will receive problem set assignments after finishing each chapter. The problem sets must be solved by handwriting and submitted to the instructor as a single PDF file.
- (3) The graphical figures must be drawn by hand. This policy is intended to enhance students' summarization and graphical description skills.
- (4) The content of submitted PDF files must be clear and easy to read. Any vague content in terms of light and dark contrast, color, etc., will not be assessed, resulting in 0 point for the related criteria. The use of a scanner is recommended over the use of a cell phone camera.
- (5) Any late submission after the due date or time will not be accepted, resulting in 0 point for the related criteria.

### 5.2.4 Attitude and In-class Participation (10 points)

- (1) Students will start with 0 points in the Attitude and In-class Participation criteria. Students may earn points through positive behaviors and actions, while negative actions and behaviors will result in point deductions.
- (2) Attitude will be evaluated as follows: Positive contributions to the learning environment may be rewarded with +0.5 points. Behaviors that disrupt the learning environment will be penalized -0.5 points per occurrence (e.g., unauthorized use of a non-English language, studying for other classes, unauthorized chatting, taking a nap, etc.). Unauthorized use of cell phones and computers will result in a -2 point deduction per occurrence. Unauthorized consumption of food and beverages is not permitted in the classroom.
- (3) In-class participation will be evaluated as follow: Active participation and communication during class may result in earning positive points. Low participation and weak involvement in discussion will lead to the point deductions.
- (4) The assessment of the Attitude and In-class Participation criteria will be subject to the instructor's discretion.

### 5.2.5 Attendance (5 points)

- (1) Attendance will be recorded and graded as follow: Students will receive 5 points for no unexcused absence and no late arrival. Being late will be penalized by -0.5 point per each class. One unexcused absence will be penalized by 1 point. Two unexcused absences will be penalized by 3 points. Three unexcused absences will be penalized by 5 points and will not receive any points in Attendance criterion. Four or more unexcused absences will be result in FAIL grade of the course.

## 6. Criteria for Awarding Credits

| A (100.00 ~ 75.00) |       |       | B (74.99 ~ 60.00) |       |       | C (59.99 ~ 45.00) |       |       | D (44.99 ~ 30.00) |       |       | F     |
|--------------------|-------|-------|-------------------|-------|-------|-------------------|-------|-------|-------------------|-------|-------|-------|
| A+                 | A0    | A-    | B+                | B0    | B-    | C+                | C0    | C-    | D+                | D0    | D-    |       |
| 100.00             | 89.99 | 79.99 | 74.99             | 69.99 | 64.99 | 59.99             | 54.99 | 49.99 | 44.99             | 39.99 | 34.99 | 29.99 |
| ~                  | ~     | ~     | ~                 | ~     | ~     | ~                 | ~     | ~     | ~                 | ~     | ~     | ~     |
| 90.00              | 80.00 | 75.00 | 70.00             | 65.00 | 60.00 | 55.00             | 50.00 | 45.00 | 40.00             | 35.00 | 30.00 | 0     |

- (1) The course will follow an absolute grading (criterion-referenced grading) system. Grades will be assigned based on the over evaluation points that each student achieves.
- (2) Students with outstanding academic achievement may be recommended for the Academic Excellence Award and the Principal's Award, and their performance is recorded under the core competencies in the Subject Details. However, students with poor academic attitude or poor attendance will be excluded from consideration regardless of the achievement.

- Attendance for at least three-quarters of the total class hours is required to earn credit. An excessive number of excused or medical absences may result in a FAIL grade for the course, in accordance with the Korea Science Academy Regulations.
- For PASS/FAIL courses, if the instructor determines that a specific evaluation component is insufficient, an additional assessment may be conducted.
- The method for calculating the baseline score used to award credit to students with absences will be determined by the committee within the Department of Chemistry and Biology at the Korea Science Academy.
- If the baseline score cannot be calculated, the grading method will be determined by the committee within the Department of Chemistry and Biology at the Korea Science Academy.

## 7. Weekly Plans

| Week (period)                | Topic and Contents                                                                                                                                                                                 | Notes                                              |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Week 1</b><br>2.23.~3.1.  | Chapter 1. Structure and Bonding; Acids and Bases                                                                                                                                                  |                                                    |
| <b>Core Competencies</b>     | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                                    |
| <b>Week 2</b><br>3.2.~3.8.   | Chapter 2. Alkanes: The Nature of Organic Compounds                                                                                                                                                | <b>3.2.</b><br><b>Substitute</b><br><b>Holiday</b> |
| <b>Core Competencies</b>     | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                                    |
| <b>Week 3</b><br>3.9.~3.15.  | Chapter 3. Alkenes and Alkynes: The Nature of Organic Reactions                                                                                                                                    |                                                    |
| <b>Core Competencies</b>     | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                                    |
| <b>Week 4</b><br>3.16.~3.22. | Chapter 4. Reactions of Alkenes and Alkynes                                                                                                                                                        |                                                    |
| <b>Core Competencies</b>     | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                                    |
| <b>Week 5</b><br>3.23.~3.29. | Chapter 5. Aromatic Compounds                                                                                                                                                                      |                                                    |
| <b>Core Competencies</b>     | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                                    |
| <b>Week 6</b><br>3.30.~4.5.  | Chapter 6. Stereochemistry at Tetrahedral Centers                                                                                                                                                  |                                                    |
| <b>Core Competencies</b>     | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                                    |

|                              |                                                                                                                                                                                                    |                                        |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Week 7</b><br>4.6.~4.12.  | Chapter 7. Organohalides: Nucleophilic Substitutions and Eliminations                                                                                                                              |                                        |
| <b>Core Competencies</b>     | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                        |
| <b>Week 8</b><br>4.13.~4.19. | <b>Midterm Exams (4.14.~4.17.)</b>                                                                                                                                                                 | <b>4.13.</b><br><b>Regular Classes</b> |

|                               |                                                                                                                                                                                                    |                                           |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Week 9</b><br>4.20.~4.26.  | Computational Chemistry Individual Project Introduction                                                                                                                                            |                                           |
| <b>Core Competencies</b>      | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                           |
| <b>Week 10</b><br>4.27.~5.3.  | Chapter 8. Alcohols, Phenols, Ethers, and Their Sulfur Analogs                                                                                                                                     |                                           |
| <b>Core Competencies</b>      | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                           |
| <b>Week 11</b><br>5.4.~5.10.  | Chapter 9. Aldehydes and Ketones: Nucleophilic Addition Reactions                                                                                                                                  | <b>5.5.</b><br><b>Children's Day</b>      |
| <b>Core Competencies</b>      | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                           |
| <b>Week 12</b><br>5.11.~5.17. | Chapter 10. Carboxylic Acids and Derivatives: Nucleophilic Acyl Substitution Reactions                                                                                                             |                                           |
| <b>Core Competencies</b>      | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                           |
| <b>Week 13</b><br>5.18.~5.24. | Chapter 11. Carbonyl Alpha-Substitution Reactions and Condensation Reactions                                                                                                                       |                                           |
| <b>Core Competencies</b>      | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                           |
| <b>Week 14</b><br>5.25.~5.31. | Chapter 12. Amines                                                                                                                                                                                 | <b>5.25.</b><br><b>Substitute Holiday</b> |
| <b>Core Competencies</b>      | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical Thinking Skills / Creativity / Ability of Collaborate |                                           |
| <b>Week 15</b><br>6.1.~6.7.   | Chapter 13. Structure Determination (tentative)<br>Computational Chemistry Individual Project Presentation                                                                                         | <b>6.3.</b><br><b>Local Election Day</b>  |
| <b>Core Competencies</b>      | Analytical and Creative Thinking / Complex Communication / Integrity and Ethical Decision Making / Complex Problem-solving Skills / Critical                                                       |                                           |

|                               |                                                       |   |
|-------------------------------|-------------------------------------------------------|---|
|                               | Thinking Skills / Creativity / Ability of Collaborate |   |
| <b>Week 16</b><br>6.8.~6.14.  | <b>Make-up class week</b>                             | - |
| <b>Week 17</b><br>6.15.~6.19. | <b>Final Exams (6.18.~6.19.)</b>                      | - |

## 8. Note

The course plan and policies may change at the instructor's consideration, depending on the given circumstances.